PROBLEM STATEMENT · SUSTAINABILITY

Leaf*It*

Redesign your block. Leave it greener.
A Gemini planning agent on a satellite-built digital twin of your neighbourhood: it tests where trees, cool roofs and shaded bus stops cool people most, shows what each option costs, and turns the plan into a funding proposal and vendor quotes.

Gemini · Vertex AI | Function calling + vision | Earth Engine | Google Maps Platform | Cloud Run

Friendly Neighbourhood Devs

Cities are heating up, and the money to cool them is spent *blind*.

20 °C+

Heat is hyper-local

Sunlit pavement can be 20 °C hotter than the same street in tree shade. City-wide averages hide the blocks where people actually suffer: bus stops, markets, dense homes.

?

No way to test a plan

Residents' associations, NGOs and ward officers choose between trees, cool roofs and shade by guesswork. Existing tools are dashboards: they show the heat, not which fix works where, for whom, at what cost.

→

Plans stall before action

Even a good idea needs a costed plan, a funding pitch and local suppliers. Each is a separate manual job, so most plans never reach the street.

From a satellite map to a *funded, quoted* plan in one flow.

01 · FIND

Draw your area

Search a place on a Google satellite map and drag a box (200 m to 1 km²).

02 · TWIN

Real data, in seconds

Surface heat, every building and its height, existing tree canopy, roads, residents and climate, built into a 2.5D twin.

03 · PLAN

Design with Gemini

Place items by hand with live numbers, or brief the Gemini agent in plain words. It tests options and you browse and compare them all.

04 · ACT

Fund and build

A bill of materials by species, a Gemini-drafted funding proposal, and real nurseries and contractors within 25 km with a ready quote message.

Budget-aware throughout: one budget, a live "used / left" bar, and nothing can be placed or planned that the money can't pay for.

A digital twin built from *nine* real sources, not a mock-up.

Google Earth Engine

- Landsat 8/9 surface heat, hot-season median of 3 years
- ESA WorldCover 10 m land cover
- Open Buildings v3 footprints + 2.5D heights
- Meta/WRI 1 m tree-canopy height
- WorldPop residents by age (children, over-65s)

Google Maps Platform

- Places: landmarks (markets, schools, clinics, bus stops) and real vendors
- Air Quality API: today's AQI
- Map Tiles: satellite ground of the 3D twin

Open data

- OpenStreetMap roads and carriageway widths
- Open-Meteo weather + humidity normals (climate class)

A model that respects what is already there

- Each new tree adds only what existing shade hasn't already done, including canopy in a 150 m ring around the box
- Species grow in over years; surface, air and "where vulnerable people live" cooling are reported separately
- Humid climates get an airflow spacing rule
- Every coefficient is cited on an in-app Method page

Gemini *decides*. The simulator *calculates*. Code *checks*.

1 · See
Brief in plain words + budget, a place brief (zones, hotspots, canopy, residents, surroundings, climate) and two labelled maps: heat and canopy

→

2 · Strategise
Calls `run_scenario` 2–4 times with different levers: goal, zones, landmarks, species, roofs for homes, bus shelters

→

3 · Self-critique
`check_plan`: budget used, trees wasted in shade, hot or vulnerable zones left out, species mix

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4 · Revise
Choosing a plan with >10% of trees in existing shade is refused in code; Gemini must fix it (up to 2 rounds)

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5 · Explain
Structured JSON answer: headline, trade-offs, zone-by-zone actions, caveats

Gemini (function calling, vision, thinking) deterministic code: optimiser, simulator, checks

Numbers guardrail

Every figure in Gemini's prose is matched against tool outputs (allowing ₹ lakh/crore, kg⇔t, % rescaling). Anything unmatched is shown to the user as "not from the simulator".

Weighs what can't be scored

Trees cool streets, cool roofs cool the rooms below, shelters shade people waiting. No single number ranks them; Gemini weighs them against the brief.

Honest by design

No hidden fallback: if Gemini is unavailable the planner says so instead of inventing a plan. Every option tested is returned and browsable.

Two Gemini workflows, each *grounded* in the simulator.

1 · Planning agent (Studio → "Ask AI to plan")

- Multimodal: text brief + place brief JSON + two generated map images with the same 3×3 zone grid as the tools
- 5 function tools, up to 16 turns, configurable thinking budget
- Returns every option it tested with what goes where: tree species per zone, roofs, shelters, nearby roads and landmarks
- Apply Gemini's pick or any other option; all of them land in Plan Options to compare side by side
- "About this plan" card stays reachable on the map, with Show-zone highlighting

2 · Funding-proposal writer (Review → "Write the proposal")

- Drafts a one-page proposal for a municipal officer or CSR committee from the saved plan only
- Place-aware: climate, existing canopy, species chosen, residents and vulnerable residents who benefit
- Money is pre-formatted in the reader's currency (14 currencies) so the model never does arithmetic
- Same numbers guardrail; the draft header says if any figure needs checking

Runs on Vertex AI

On Cloud Run, Gemini is called through Vertex AI as the service account: no API key in the deployment. Rate-limit replies are waited out, not failed.

Different options serve *different people*.

OPTION TESTED	MIX	STREET COOLING	HOTTEST SPOT	STILL TOO HOT	CO_2 / YR	WHO ELSE BENEFITS
Whole-area cooling	666 trees · 5 roofs	**−0.29 °C**	−1.9 °C	17%	**9.4 t**	cheapest per degree (₹68 L/°C)
Hottest ground first	666 trees	−0.21 °C	**−2.4 °C**	**13%**	8.0 t	the hottest outdoor spots
Homes and bus stops	233 trees · 51 roofs · 1 shelter	−0.11 °C	−1.8 °C	21%	3.4 t	**620 residents** under cool roofs, −0.18 °C where vulnerable people live, 1 shaded bus stop

A single "most cooling" number would always pick the first row. LeafIt's agent picks the row that matches the brief and says why. Every figure above is from the simulator; species are matched to Delhi's hot-dry climate (Neem, Arjun, Amaltas, Indian coral tree).

Residents & RWAs
a costed plan for their own block

Municipal officers
cooling per rupee, by zone

NGOs & CSR teams
a sourced funding proposal

Local vendors
itemised quote requests

Beyond a chatbot on a dashboard.

An agent with levers, not a prompt

Gemini operates a real optimiser and simulator through function calls, the way a planner would use a spreadsheet, and must defend its choice.

Self-critique enforced in code

The agent's own check report can block its choice. The revision loop is real: a wasteful plan cannot be recommended.

It sees the place

Generated heat and canopy maps, labelled with the tools' zone grid, let Gemini reason about "the hot north-east corner beside the market".

Existing shade counts

The twin knows today's trees from 1 m canopy height, so new trees go where they add shade, not where shade already is.

Equity built in

WorldPop age data turns into "cooling where children and over-65s live", a number the agent can plan for.

Plan → money → people

The same session ends in a funding proposal and real vendors with a WhatsApp-ready quote.

BUILT PROPERLY

TECHNICAL · UX

It works today, on *Google Cloud*.

Live

Cloud Run + Vertex AI

One container serves API + web app. Cloud Build → Artifact Registry → Cloud Run in one script. No keys in the deployment: Gemini and Earth Engine use the service account.

156

Automated tests

113 backend + 43 frontend, all offline: simulator physics, the agent loop with a scripted Gemini, the numbers guardrail, plan → BOM → vendors. One shared model file keeps browser and server simulators identical.

Fast

Instant feedback

The simulator runs in a Web Worker on every placement; the 3D scene renders only when something changes. Built areas are cached.

Usable and accessible

Plain-language AI brief, live budget used/left, keyboard shortcuts, ARIA labels and focus states, light and dark themes, 14 currencies.

Every screen, every size

Tested from 375 px phones to 3440 px ultrawide: no sideways scrolling, content centred, a sheet-style UI on phones.

From one block to every ward.

Reach

Works wherever Google Open Buildings has data: Africa, South and Southeast Asia, Latin America, which covers many of the world's fastest-heating cities. Prices, budget and currency are editable for any market.

Scales on Google Cloud

Stateless, autoscaling Cloud Run; Vertex AI with no free-tier limits; the area and plan cache moves to Firestore / Cloud Storage by re-implementing one repository layer.

Next

- Ward-level heat-priority maps for municipalities
- More interventions: green walls, permeable paving, water
- Track planted trees against the plan with fresh imagery
- Verified vendor marketplace

Sustainable to run: the heavy maths runs in the user's browser; the server only builds areas, saves plans and calls Gemini.

TRY IT

Leave your block *greener*.

Live prototype

`leafit.cloud.run`

3-minute demo

`{{VIDEO_URL}}`

Category: Sustainability · Friendly Neighbourhood Devs